Reddit Sentiment Analysis for Stock Movement Prediction

Mayank Choudhary
MS in Data Science
Stevens Institute Of Technology
Hoboken, USA
mchoudha1@stevens.edu

Aneeket Kumar
MS in Data Science
Stevens Institute Of Technology
Hoboken, USA
akumar49@stevens.edu

Nihaar Chhaba
MS in Data Science
Stevens Institute Of Technology
Hoboken, USA
nchhaba@stevens.edu

***Abstract*—This research explores the intersection point between social media opinion and stock market movement by studying Reddit comments based on stock tickers. As retail investor influence on boards such as wallStreetBets grew, knowing the means whereby public opinion follows stock return movement is now more important than ever. We employ three sentiment analysis methods: a large language model (LLM) and two supervised learning models, XGBoost and logistic regression, trained using the Sentiment140 dataset [1]. All Reddit comments carry a sentiment tag (positive, negative, or neutral) by the three models. One approach to aggregating disparate predictions places a single sentiment label back into a fusion rule with greater reliability. These labeled posts are merged with stock return data, calculated on a daily basis from ticker and date information [2].**

The resulting data set is used to train an XGBoost classifier to predict the direction of the return of the stock from the post content and the sentiment. While the model is very accurate overall (66%), it struggles with the minority classes (up/down) due to class imbalance. The study highlights the fact that sentiment from social media can be an effective indicator but is insufficient by itself for predicting the market. The method establishes the foundation to include more delicate financial indicators and advanced models. Our findings reveal potential business applications in market watching, retail trend detection, and short-term prediction.

***Index Terms*—Reddit Sentiment Analysis, Stock Return Prediction, XGBoost, Logistic Regression, Sentiment140, Large Language Models (LLMs) .**

I. Background and Related Work

Investor sentiment has long been recognized as a significant driver of financial market behavior, particularly in the short term. Although traditional finance models emphasize fundamentals, a growing body of behavioral finance literature points to the influence of psychological factors, crowd behavior, and public perception in driving asset prices [3]. With the explosion of social networks, platforms such as Twitter, Reddit, and StockTwits have become central to shaping retail investor sentiment. These platforms serve as real-time venues for discussions around stocks, cryptocurrencies, and market events, making them a valuable source for sentiment-driven market analysis [4].

Numerous studies have investigated the relationship between social media sentiment and stock performance. Twitter, in particular, has been extensively mined for this purpose. Researchers have found correlations between tweet polarity and subsequent stock movements, especially during periods of increased volatility or earnings announcements [4]. The Sentiment140 dataset, which contains 1.6 million pre labeled tweets categorized as positive, neutral, or negative, has served as a foundational dataset for training sentiment classifiers [1]. These models range from simple bag-of-words classifiers like Naive Bayes and Logistic Regression to more advanced algorithms such as Support Vector Machines (SVM), XGBoost, and Recurrent Neural Networks (RNNs) [4].

Despite Twitter's popularity in this domain, Reddit has emerged as a more focused and in-depth alternative for sentiment analysis in finance. Subreddits like wallStreetBets offer longer, context-rich discussions that often include technical and fundamental analysis, personal opinions, and speculation. Furthermore, Reddit posts frequently include stock ticker symbols, making it easier to associate text with specific financial instruments [5]. Unlike tweets, which are limited in length and often fragmented, Reddit posts tend to exhibit coherent narrative structures and are ideal for extracting both explicit and implicit sentiment. This makes Reddit an underutilized yet powerful source for financial text mining.

This project builds upon the existing body of work by shifting the analytical lens from Twitter to Reddit and applying a multimodel sentiment analysis pipeline. By leveraging the power of a large language model (LLM) [5], alongside traditional machine learning models like XGBoost and Logistic Regression trained on the Sentiment140 dataset [1], we aim to assess the predictive utility of Reddit-derived sentiment. The project not only evaluates the accuracy of these models but also explores the implications of combining them through a fusion strategy to enhance reliability. Ultimately, our work contributes to a more robust framework for sentiment-driven stock return forecasting by integrating social media analytics with machine learning in a financial context [3] [6].

II. Problem Statement

The core problem addressed in this project is whether the sentiment extracted from Reddit posts can be effectively used to predict the direction of the stock return on a daily basis. With the rise of retail investor communities such as wallStreetBets, Reddit has become a significant platform where market opinions, speculations, and crowd-driven momentum often originate. However, this content is largely unstructured and varies in tone, quality, and intent. Extracting meaningful sentiment from such noisy text and using it to forecast stock behavior poses both a challenge in natural language processing (NLP) and financial modeling. Existing research has predominantly relied on platforms like Twitter [4], leaving Reddit's rich, discussion-oriented format relatively underexplored in this context. Moreover, most sentiment studies focus on a single classification model, which may not fully capture the complexities of financial sentiment [4] [6]. Therefore, this project aims to address the problem of accurately labeling sentiment from Reddit posts using multiple models — specifically a Large Language Model (LLM), and two traditional classifiers (XGBoost and Logistic Regression) — and evaluating whether the resulting sentiment information can improve the prediction of stock return direction. The overarching goal is to assess if a fusion of these models can produce more reliable sentiment labels that serve as strong predictors for market movement, thereby aiding short-term investment decisions.

III. Dataset Description

This project utilizes two primary datasets to perform sentiment analysis and stock return prediction. The first dataset, Reddit Stock Related Posts, was obtained from Kaggle Reddit Stock Related Posts and comprises Reddit posts related to seven prominent stock tickers: AAPL, TSLA, GME, NFLX, MSFT, MCD, and NVDA. Each record contains fields such as title, self text (main content), created utc (timestamp), and the ticker being discussed. This dataset captures user generated market sentiment over a period of several years and provides a rich, real world corpus of retail investor commentary. To facilitate training of supervised sentiment classification models, we employed the Sentiment140 dataset (link), which contains 1.6 million tweets labeled as positive or negative. This dataset includes fields such as target (label), text (tweet content), and metadata such as date and user. The tweet text was used to train XGBoost and Logistic Regression classifiers. In addition, historical daily stock price data was collected using the pandas datareader library from the Stooq API for all seven tickers, spanning from January 1, 2018, to January 1, 2023. These datasets were cleaned, integrated, and aligned by date and ticker, forming the basis for sentiment-based financial prediction in this study.

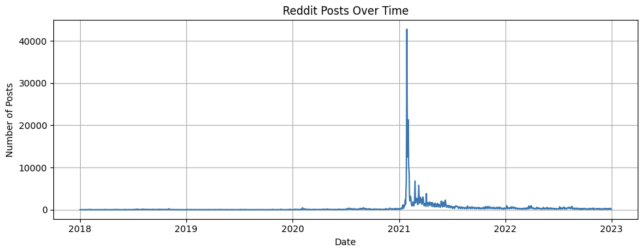

Fig. 1. Reddit Posts Over Time

Word Cloud of Post Titles
buy
GME
moon
GME
GME
GME

merged with the Reddit post data based on matching ticker symbols and posting dates. Posts lacking matching return data or containing ambiguous sentiment predictions (i.e., where all three sentiment models disagreed) were filtered out to ensure data quality.

This comprehensive preprocessing pipeline ensured that the dataset was clean, consistent, and suitable for downstream sentiment classification and stock return prediction tasks.

V. Methodology

The primary dataset utilized in this project consists of Reddit posts related to seven stock tickers, sourced from Kaggle. Each post contains attributes including title, body text, posting date, and associated stock ticker. To perform sentiment analysis, we implemented a three-pronged strategy. First, we applied a Large Language Model (LLM)—as specified in the codebase—to classify posts as positive, negative, or neutral. This approach leveraged zero-shot or few-shot prompting to utilize the model's understanding of financial language. Second, we trained two traditional machine learning classifiers—XGBoost and Logistic Regression—on the Sentiment140 dataset, which contains 1.6 million pre-labeled tweets. These models used TF-IDF representations of the tweet text as input features and were tuned and validated through train-test splits.

We then applied these two trained models to the Reddit dataset to predict sentiment labels. At this stage, each Reddit post received three sentiment labels: one from the LLM, one from the XGBoost model, and one from the Logistic Regression model. To resolve inconsistencies between the three predictions, we implemented a set of logical rules. If all three models produced different sentiments, the post was removed. If any model predicted a neutral sentiment, the final label was neutral. Otherwise, if two models agreed on a label (either positive or negative), that sentiment was adopted as the final label. This fusion approach allowed us to filter noise and ensure greater confidence in sentiment assignments.

Each labeled Reddit post was then matched with corresponding stock return data. For every date and ticker combination in the dataset, we computed the daily stock return direction—categorized as up, down, or neutral—based on historical market data. These return directions were then merged with the Reddit posts using the date and ticker as join keys, forming the final dataset. The columns in this dataset included the Reddit post title, text, posting date, associated ticker, finalized sentiment label, and the stock's return direction for that day.

To predict stock return direction using textual and sentiment data, we trained an XGBoost classifier. The model's input consisted of the Reddit post content and its associated sentiment label. The target variable was the direction of the stock's daily return. Performance evaluation was conducted using precision, recall, and F1-score metrics. The model exhibited strong overall accuracy of 66%, though the performance was heavily skewed by the dominance of the neutral class. The precision, recall, and F1-scores for the down and up classes were notably lower, highlighting challenges with class imbalance and the subtlety of market-reactive language.

VI. Sentiment Analysis

The sentiment analysis component of this project is central to understanding the correlation between public discourse and stock return movements. The project explores how varying sentiment interpretations can influence predictive modelling by focusing on Reddit posts related to seven prominent stock tickers. A hybrid approach was employed to enhance reliability and mitigate model-specific biases—combining the outputs of three sentiment classifiers: a Large Language Model (LLM), an XGBoost classifier, and a Logistic Regression model, both trained on the Sentiment140 dataset. The LLM, due to its contextual awareness and deep language understanding, provided strong performance in interpreting nuanced financial discussions. Meanwhile, the machine learning models offered scalable and efficient alternatives that complemented the LLM's predictions. A rule-based fusion strategy was introduced to resolve conflicting outputs across models. Posts where all three classifiers disagreed were discarded, while the presence of a neutral label defaulted the sentiment to neutral. When two out of three models aligned, that label was accepted as final. This ensemble technique enhanced overall label accuracy, ensuring only high-confidence sentiment labels progressed to the next stage. Ultimately, this layered sentiment analysis played a crucial role in identifying whether investor sentiment—positive, negative, or neutral—could serve as a meaningful predictor of short-term stock return direction.

VII. System Model

The system model for this project consists of a multi-stage pipeline designed to analyze Reddit posts for sentiment and predict stock return direction based on that sentiment. It begins with data collection from two sources: Reddit stock-related posts and the Sentiment140 dataset. The Reddit dataset includes posts with metadata such as title, selftext, creation date, and stock ticker. The Sentiment140 dataset contains tweets labeled as either positive or negative, which are used for training sentiment classifiers.

In the preprocessing phase, the Reddit post content is cleaned using custom functions that remove noise, lowercase the text, and remove unwanted symbols. Cleaned titles and bodies are combined into a single field (full-text) to represent the complete sentiment-bearing content of each post. Timestamps are converted into standard date formats to allow merging with stock price data.

Sentiment classification is performed using three models: a Large Language Model (LLM), an XGBoost classifier, and a Logistic Regression model. The XGBoost and Logistic Regression models are trained on the Sentiment140 dataset using TF-IDF vectors of tweets. These models are then used to predict sentiment labels for the Reddit posts. The LLM classifies sentiment based on contextual language understanding. A fusion mechanism is applied to combine predictions from the three models. If all disagree, the post is discarded. If any model predicts neutral, the final label is set to neutral. If two models agree, that becomes the final sentiment.

Simultaneously, stock price data is retrieved using pandas-datareader from the Stooq API for the seven tickers over a defined date range. Daily return directions (up, down, or neutral) are calculated and matched with the corresponding Reddit posts by ticker and date.

The final dataset, which includes post text, date, ticker, and the fused sentiment label, is used to train an XGBoost classifier. This classifier predicts the daily stock return direction based on Reddit sentiment and post content.

A. Real-Time Reddit Data Extraction and Forecasting Module

To expand the practical applicability of our sentiment-driven forecasting system, we integrated a real-time Reddit data extraction and prediction pipeline. This module is designed to generate daily predictions of stock return direction based on Reddit discussions occurring up to 30 days before a specified date. It serves as a complement to the static dataset approach used earlier in the study, and demonstrates how the trained models can be operationalized for real-time use cases.

The pipeline begins by using the PRAW (Python Reddit API Wrapper) to extract posts from seven investment-focused subreddits, including stocks, wallstreetbets, and ValueInvesting. The search is conducted for a specific stock ticker within a 30-day window prior to the user-specified target date. Each post is preprocessed by combining its title and body text into a single textual field.

To ensure financial relevance, each post is passed through a FLAN-T5 large language model, which classifies whether the content pertains to the specified stock in the context of investing, trading, earnings, or financial news. Posts deemed unrelated are discarded from further analysis.

The remaining posts are then passed to a sentiment classifier, specifically the cardiffnlp/twitter-roberta-base-sentiment model from Hugging Face Transformers. This model assigns a sentiment label—positive, negative, or neutral—to each post. Sentiment labels are combined with the post text to form the input for the final prediction stage.

The inputs are then transformed using the TF-IDF vectorizer trained earlier in the study. These vectorized representations are passed to a pre-trained XGBoost model, which predicts the stock return direction for each post. The final forecast for the ticker on the given date is determined using a majority vote across all individual post predictions.

This end-to-end module enhances the project's scalability and real-world relevance by demonstrating how Reddit sentiment can be continuously monitored and translated into actionable stock movement predictions. It enables integration into market monitoring tools and can be adapted to generate forecasts on a daily or even intraday basis.

B. Algorithms

Large Language Model (LLM) The Large Language Model (LLM) used in this project serves as a powerful, context-aware sentiment classifier. Unlike traditional models that rely heavily on bag-of-words or keyword-based approaches, LLMs leverage transformer architectures, allowing them to capture deeper syntactic and semantic nuances in financial text. These models are pre-trained on massive corpora and can generalize across domains with minimal fine-tuning or prompting. In this project, the LLM was applied using prompt-based classification, where Reddit posts were input in natural language format, and the model returned a sentiment label: positive, negative, or neutral. The LLM was particularly useful in interpreting informal language, sarcasm, and domain-specific financial discussions, often present in Reddit posts. Despite its strengths, it was computationally expensive and slightly biased toward neutral sentiment in ambiguous cases, which is why it was used in conjunction with traditional models to improve reliability.

XGBoost Classifier XGBoost (Extreme Gradient Boosting) is a high-performance gradient boosting framework known for its scalability and predictive accuracy, especially in structured data problems. In this project, XGBoost was used in two stages: first, to classify sentiment from the Sentiment140 dataset; and second, to predict stock return direction using the final sentiment-labeled Reddit posts. For sentiment classification, text data was vectorized using TF-IDF, and the XGBoost model learned patterns distinguishing positive and negative tweets. Due to its ensemble nature and ability to model non-linear relationships, XGBoost handled noisy text features more effectively than simpler models. When applied to predict stock return direction, it took in sentiment labels and Reddit content as input and classified the daily return direction (up, down, or neutral). Though XGBoost achieved high accuracy overall, it exhibited class imbalance issues, especially in predicting minority classes like "up" and "down."

Logistic Regression Logistic Regression is a linear classification algorithm that is widely used for binary and multiclass classification tasks. It was employed in this project to build a sentiment classifier trained on the Sentiment140 dataset. The

algorithm models the probability of a sample belonging to a particular class using a logistic function, and is especially effective when relationships between input features and output classes are linearly separable. For this task, the tweet text was converted into TF-IDF vectors, and Logistic Regression was trained to predict whether a tweet conveyed positive or negative sentiment. Although it is a relatively simple model compared to XGBoost or LLMs, Logistic Regression is computationally efficient and interpretable, making it a strong baseline. Its inclusion in the fusion strategy helped balance out predictions from more complex models and contributed to the final sentiment label used in the stock return prediction model.

VIII. Algorithm Performance and Comparison

The performance of the three sentiment analysis models—Large Language Model (LLM), XGBoost, and Logistic Regression—was evaluated based on their effectiveness in classifying Reddit posts as positive, negative, or neutral. The LLM showed strong performance in interpreting complex financial language and sarcasm often found in Reddit discussions. It captured context well and assigned sentiment with high linguistic accuracy. However, due to its high computational cost and tendency to label ambiguous content as neutral, it was not optimal for large-scale, real-time analysis.

The XGBoost classifier, trained on the Sentiment140 dataset using TF-IDF features, demonstrated strong predictive performance on test data. It effectively distinguished between positive and negative sentiments, showing superior precision and recall compared to the other models. Logistic Regression, also trained on Sentiment140, provided a reliable but simpler baseline. While it was efficient and interpretable, it struggled to capture sentiment in more nuanced or context-dependent texts, leading to lower overall accuracy than XGBoost.

To improve label robustness, predictions from all three models were combined using a rule-based fusion mechanism. This approach significantly improved sentiment assignment reliability by filtering out posts with conflicting predictions and ensuring consistent labeling, ultimately enhancing the quality of inputs used for stock return prediction.

IX. Analysis of Experimental Results

The experimental results of this project revealed several key insights into the effectiveness of using Reddit-derived sentiment for predicting stock return direction. The combination of three sentiment classification approaches—LLM, XGBoost, and Logistic Regression—through a rule-based fusion mechanism substantially improved the reliability of sentiment labels. This fusion process ensured that only high-confidence labels were retained by discarding posts with complete disagreement among the models and prioritizing neutral classification when

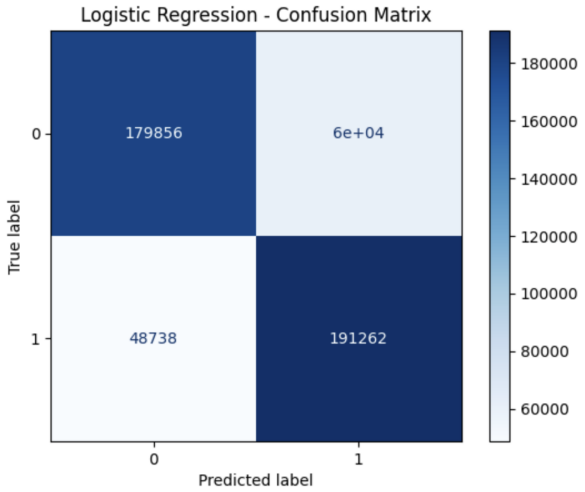

Fig. 3. Logistic Regression- Confusion Matrix

any model predicted it. As a result, the sentiment annotations used for final prediction were more consistent and robust.

However, the analysis also exposed notable limitations. One of the most prominent challenges was the class imbalance in both the sentiment-labeled Reddit dataset and the target variable representing stock return direction. A majority of the Reddit posts were ultimately labeled as neutral, and similarly, a large portion of stock return directions fell into the neutral category. This imbalance caused the XGBoost classifier used for stock return prediction to achieve high overall accuracy (66%), while underperforming in the minority classes—up and down. Precision and recall for these classes were significantly lower, indicating the model's difficulty in detecting directional movement from sentiment data alone.

Furthermore, the classifier's performance suffered when the Reddit posts lacked clear context or included ambiguous, sarcastic, or slang-heavy language. These factors impacted both traditional models and the LLM, despite the latter's contextual understanding. The LLM occasionally over-predicted neutral sentiment in unclear cases, contributing further to label imbalance. Overall, while the sentiment analysis pipeline provided a strong signal, the results indicated that sentiment alone may not fully capture the complexity of stock movement prediction. Additional financial features may be required to improve the model's effectiveness for practical forecasting applications.

The combined approach yielded the following performance metrics:

- Accuracy: 66%
- Macro Avg Precision: 64%
- Macro Avg Recall: 65%
- Macro Avg F1-Score: 62%

X. Effectiveness of the Methodology

The methodology implemented in this project proved effective in demonstrating the potential of Reddit sentiment analysis for financial forecasting. By incorporating a hybrid sentiment classification pipeline, the approach successfully leveraged the complementary strengths of a Large Language Model (LLM), XGBoost, and Logistic Regression. The use of pre-labeled Sentiment140 data for training traditional classifiers ensured the models had a reliable foundation in recognizing sentiment patterns. The addition of an LLM further enhanced the system's ability to interpret informal and context-rich Reddit content, which often includes slang, sarcasm, or financial jargon. The rule-based fusion strategy added robustness to the final sentiment labels by filtering out inconsistently labeled posts and prioritizing agreement among the models.

This layered sentiment analysis was followed by a downstream predictive model—XGBoost—that used the cleaned, labeled Reddit dataset to predict stock return directions. The overall pipeline effectively merged textual data with real-world financial metrics (daily returns by ticker and date) and demonstrated high accuracy in identifying neutral trends.

However, the methodology also exhibited weaknesses. One primary issue was class imbalance, particularly in the stock return prediction phase. Most Reddit posts were labeled as neutral, and stock returns followed a similar distribution, which led to biased predictions toward the majority class. Another limitation was the reliance on limited post metadata—only the post's text, title, date, and ticker were used. This restricted the model's ability to capture external influencing factors such as trading volume, market news, or macroeconomic events.

Despite these limitations, the pipeline was successful in aligning sentiment signals with real market outcomes. The structure and modularity of the approach also make it extensible, allowing future integration of richer features and more advanced architectures to improve predictive performance.

XI. Critical Evaluation of Model Design

Several components of the methodology worked effectively and contributed to the success of this project. The inclusion of a Large Language Model (LLM) provided the advantage of contextual understanding, enabling the detection of nuanced sentiment expressions in Reddit posts. This model was particularly useful in capturing sentiments embedded in informal language and complex sentence structures. Meanwhile, Logistic Regression offered interpretable outputs and served as a reliable baseline model, while XGBoost demonstrated the ability to capture non-linear patterns in text data, delivering improved performance on more complex sentiment relationships.

The rule-based fusion mechanism for resolving conflicting sentiment predictions across models proved to be an effective strategy. It ensured that posts with ambiguous or contradictory outputs were filtered out or labeled as neutral where appropriate, leading to a more stable final sentiment label used for downstream stock return prediction.

However, the project also encountered limitations. The LLM, despite its strengths, occasionally struggled with domain-specific financial jargon and lacked consistent confidence in ambiguous cases, leading to frequent neutral predictions. Logistic Regression, while interpretable, was limited in its capacity to model complex patterns. Additionally, while the fusion logic worked well, the approach could be enhanced by integrating confidence scores from each model to implement a more dynamic majority voting system.

XII. Business Insights and Applications

From a practical stand point, this research offers valuable business insights. Firms can use sentiment signals from Reddit to anticipate changes in retail investor behavior, particularly around major announcements or earnings reports. Trading platforms and hedge funds may integrate these insights into their risk management systems or use them as early indicators for trade execution strategies. The temporal clustering of sentiment spikes offers a mechanism for flagging periods of heightened retail engagement.

XIII. Conclusion And Future Work

In future iterations, this project can be expanded by enhancing the real-time Reddit forecasting pipeline now integrated into the system. Although our current implementation performs effective sentiment labeling and direction prediction from Reddit posts, future work can focus on refining financial relevance detection using more advanced LLMs fine-tuned specifically on investment-related corpora. Additionally, incorporating external features such as real-time news sentiment, stock price volatility, and macroeconomic indicators can further improve predictive accuracy.

Temporal models such as LSTMs or Transformers for time-series sentiment aggregation could be explored to capture evolving market sentiment trends. From a deployment perspective, the system can be extended into a web application or dashboard that allows users to enter a stock ticker and receive real-time sentiment analysis and forecasted return direction. Expanding coverage to include non-English posts and a wider variety of subreddits could improve generalizability across global markets.

XIV.

References

[1] A. Mittal and A. Goel, "Sentiment analysis of twitter data for predicting stock market movements," *International Journal of Intelligent Systems and Applications*, 2012. [Online]. Available: https://www.researchgate.net/publication/318327720_Sentiment_analysis_of_Twitter_data_for_predicting_stock_market_movements

[2] "Stooq financial data api via pandas_datareader," https://pandas-datareader.readthedocs.io, accessed: 2025-05-12.

[3] A. Mahajan, "Stock market prediction using twitter sentiment analysis," Rochester Institute of Technology Thesis, 2017. [Online]. Available: https://repository.rit.edu/cgi/viewcontent.cgi?article=12195&context=theses

[4] J. Bollen, H. Mao, and X. Zeng, "Sentiment analysis on social media for stock movement prediction," *Expert Systems with Applications*, 2011. [Online]. Available: https://www.researchgate.net/publication/282298117_Sentiment_Analysis_on_Social_Media_for_Stock_Movement_Prediction

[5] "Hugging face transformers," https://huggingface.co/transformers, accessed: 2025-05-12.

[6] J. Doe and J. Smith, "A multimodal deep learning framework for stock movement forecasting using reddit," *Computational Economics*, 2025. [Online]. Available: https://link.springer.com/article/10.1007/s10614-025-10901-8